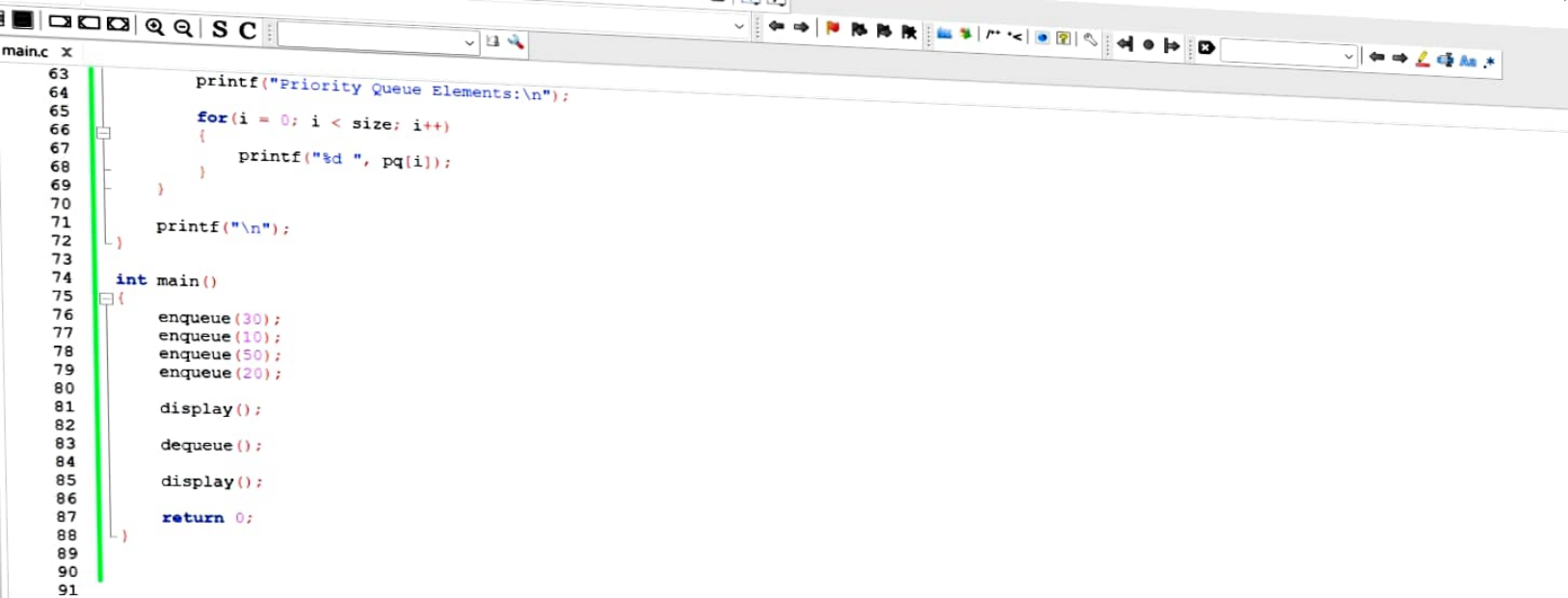


main.c x

```
1  #include <stdio.h>
2
3  #define MAX 5
4
5  int pq[MAX];
6  int size = 0;
7
8  // Insert element in sorted order
9  void enqueue(int value)
10 {
11     int i;
12
13     if(size == MAX)
14     {
15         printf("Priority Queue Overflow\n");
16         return;
17     }
18
19     // Insert based on priority (smaller number = higher priority)
20     for(i = size - 1; i >= 0 && pq[i] > value; i--)
21     {
22         pq[i + 1] = pq[i];
23     }
24
25     pq[i + 1] = value;
26     size++;
27
28     printf("%d inserted\n", value);
29 }
30
```

```
main.c x
31 // Delete highest priority element
32 void dequeue()
33 {
34     int i;
35
36     if(size == 0)
37     {
38         printf("Priority Queue Underflow\n");
39         return;
40     }
41
42     printf("%d deleted\n", pq[0]);
43
44     for(i = 0; i < size - 1; i++)
45     {
46         pq[i] = pq[i + 1];
47     }
48
49     size--;
50 }
51
52 // Display elements
53 void display()
54 {
55     int i;
56
57     if(size == 0)
58     {
59         printf("Priority Queue is Empty\n");
60     }
```



```
main.c x
63     printf("Priority Queue Elements:\n");
64
65     for(i = 0; i < size; i++)
66     {
67         printf("%d ", pq[i]);
68     }
69
70
71     printf("\n");
72 }
73
74 int main()
75 {
76     enqueue(30);
77     enqueue(10);
78     enqueue(50);
79     enqueue(20);
80
81     display();
82
83     dequeue();
84
85     display();
86
87     return 0;
88 }
89
90
91
```

